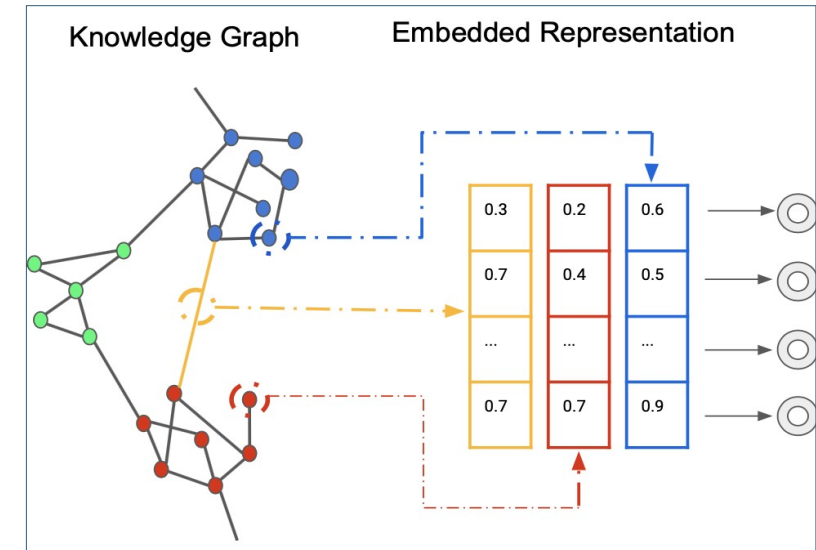


Bidirectional Monitoring of WIPO Regulation and National Legislation

A Neuro-Symbolic Approach Using AKN-XML Structured Data

Team: John Brüne, Michele Corazza, Generoso Longo, Salvatore Sapienza, Simone Vagnoni
Team Member and Supervisor: Prof. Monica Palmirani



Team

Monica Palmirani

Legal Informatics



Michele Corazza

Computational Linguistics



Generoso Longo

Computer Engineering



Salvatore Sapienza

Legal Informatics



Simone Vagnoni

Computer Engineering



John F. Brüne

Applied Statistics



Problem and Motivation

Supranational institutions must monitor and compare legislation across +190 member states to evaluate compliance and consistency.

Current bottlenecks:

- Manual comparison is extremely slow (requires legal experts for each jurisdiction)
 - Knowledge sharing across jurisdictions is limited
 - Heterogeneous legal sources make automation difficult:
 - Different legal tradition (common law, civil law)
 - Different formats (PDF, HTML, XML, scanned documents)
 - Different structures (articles, rules, paragraphs, sections)
 - Multiple languages
 - ...
- We need a **structured, explainable, scalable** method



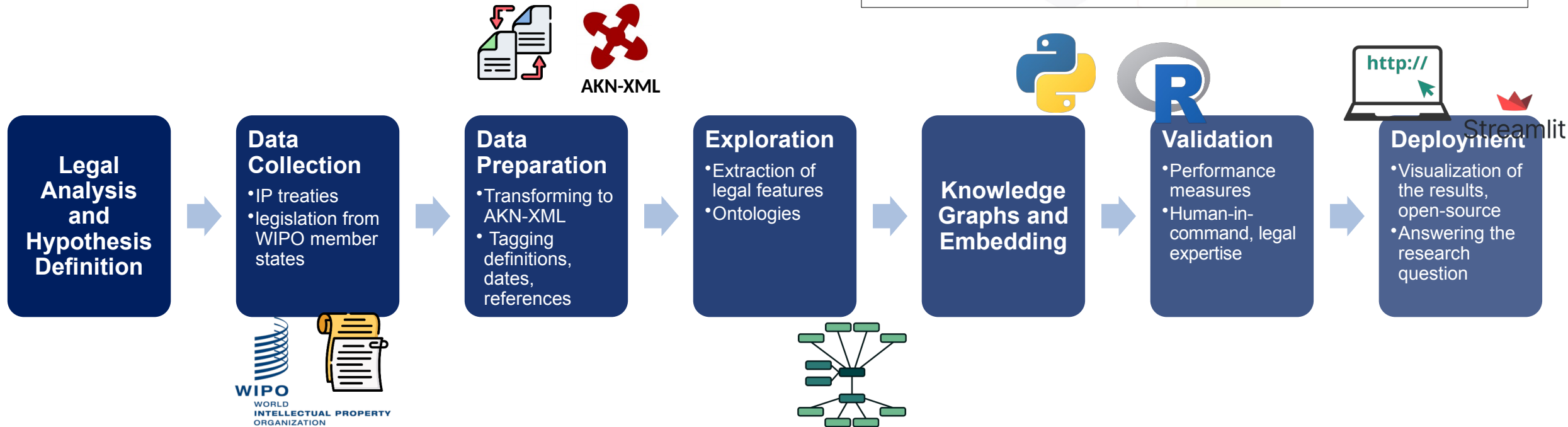
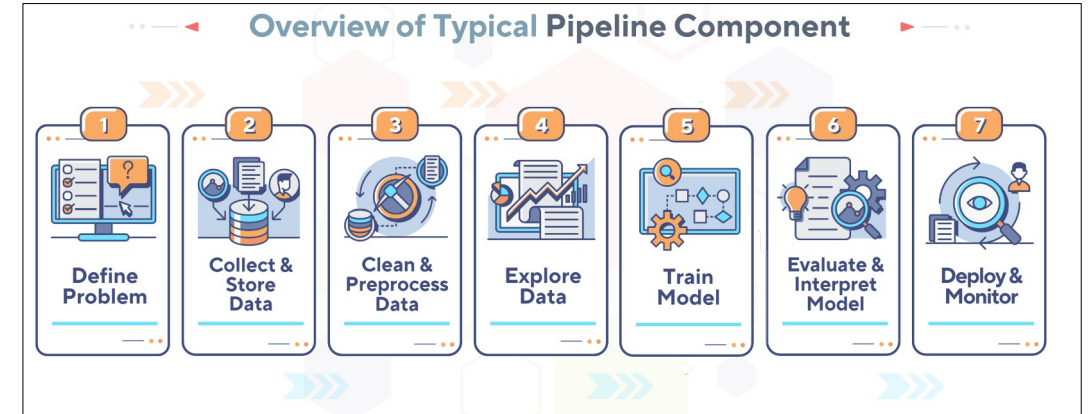
Goals

1. **Compare the WIPO regulation with Italian, German, and Swiss national legislation using Akoma Ntoso**
2. **Extract legal knowledge from the text and model it as a Knowledge Graph**
3. **Visualise the similarities and the discrepancies**
4. **Temporal modelling of the provisions**

Methodology

Legal Data Science Pipeline inspired by *Hybrid Artificial Intelligence Methodology for Legal Analysis (HAIMLA)*

(2024, Palmirani, Sapienza and Ashley)



THANK YOU FOR LISTENING!



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

PHD PROGRAMME

Law, Science and Technology

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Back-up-Slides



Mission Events Blog Network



Participation Requirements for Law As Code Hackathon

Open Problem Submission

We invite interested parties but also potential participants to submit practical problems from the judiciary and public or private sector administration or consulting that can be addressed through digital solutions.

Interdisciplinary Teams

Teams should include experts from legal and technological backgrounds to ensure comprehensive solutions.

Prototype Requirement

Teams are required to develop an executable prototype using traditional coding methods, no-code platforms, or AI-assisted tools. Appropriate documentation for such solutions is also required.

Open-Source Commitment

All solutions must shall be based on open-source tools and shall be released under an approved open-source license, ensuring accessibility and transparency.

Commitment to Publish

Teams must submit research-style documentation of their projects, detailing the problem, research, design, technology, code, and feedback from stakeholders.

Example – Akoma Ntoso

Akoma Ntoso

文A 5 Sprachen ▾

[Artikel](#)
[Diskussion](#)
[Lesen](#)
[Bearbeiten](#)
[Quelltext bearbeiten](#)
[Versionsgeschichte](#)
[Werkzeuge](#)

Akoma Ntoso (Architecture for *Knowledge-Oriented Management of African Normative Texts* using Open Standards and Ontologies) ist ein **XML-Standard** für Dokumente in den Bereichen **parlamentarische Arbeit**, **Gesetzgebung** (**Legislative**) und **Rechtsprechung** (**Judikative**).^[1]

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        <RDBURI value="/aks/wipo/act/treaty"/>
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        <RDBURI value="/aks/wipo/act/treaty/engq"/>
        <RDBDate date="1990-01-01" name="expression"/>
        <RDBLanguage language="eng"/>
      </RDBExpression>
    </identification>
  </meta>
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    <heading id="art_1_heading">Establishment of the Union; Scope of Industrial Property2</heading>
    <paragraph id="art_1_para_1">
      <sum>(1)</sum>
      <content id="art_1_para_1_content">
        <p>The countries to which this Convention applies constitute a Union for the protection of industrial property.</p>
      </content>
    </paragraph>
    <paragraph id="art_1_para_2">
      <sum>(2)</sum>
      <content id="art_1_para_2_content">
        <p>The protection of Industrial Property has as its object patents, utility models, industrial designs, trademarks, service marks, trade names, indications of source or appellations of origin, and the repression of unfair competition.</p>
      </content>
    </paragraph>
  </article>
</act>
```


Example – Akoma Ntoso

← ↻ ⓘ File /Users/johnbrune/Documents/wipo-similarity/data/wipo/html_clean/1883_Paris_clean.html#a4_C_2 ☆ ⚙ ⌵ 👤 ⋮ Chat

Article 6ter
Marks: Prohibitions concerning State Emblems, Official Hallmarks, and Emblems of Intergovernmental Organizations

(1)

(a) The countries of the Union agree to refuse or to invalidate the registration, and to prohibit by appropriate measures the use, without authorization by the competent authorities, either as trademarks or as elements of trademarks, of armorial bearings, flags, and other State emblems, of the countries of the Union, official signs and hallmarks indicating control and warranty adopted by them, and any imitation from a heraldic point of view.

(b) The provisions of [subparagraph \(a\)](#), above, shall apply equally to armorial bearings, flags, other emblems, abbreviations, and names, of international intergovernmental organizations of which one or more countries of the Union are members, with the exception of armorial bearings, flags, other emblems, abbreviations, and names, that are already the subject of international agreements in force, intended to ensure their protection.

(c) No country of the Union shall be required to apply the provisions of [subparagraph \(b\)](#), above, to the prejudice of the owners of rights acquired in good faith before the entry into force, in that country, of this Convention. The countries of the Union shall not be required to apply the said provisions when the use or registration referred to in [subparagraph \(a\)](#), above, is not of such a nature as to suggest to the public that a connection exists between the organization concerned and the armorial bearings, flags, emblems, abbreviations, and names, or if such use or registration is probably not of such a nature as to mislead the public as to the existence of a connection between the user and the organization.

(2) Prohibition of the use of official signs and hallmarks indicating control and warranty shall apply solely in cases where the marks in which they are incorporated are intended to be used on goods of the same or a similar kind.

(3)

(a) For the application of these provisions, the countries of the Union agree to communicate reciprocally, through the intermediary of the International Bureau, the list of State emblems, and official signs and hallmarks indicating control and warranty, which they desire, or may hereafter desire, to place wholly or within certain limits under the protection of this Article, and all subsequent modifications of such list. Each country of the Union shall in due course make available to the public the lists so communicated. Nevertheless such communication is not obligatory in respect of flags of States.

(b) The provisions of [subparagraph \(b\) of paragraph \(1\)](#) of this Article shall apply only to such armorial bearings, flags, other emblems, abbreviations, and names, of international intergovernmental organizations as the latter have communicated to the countries of the Union through the intermediary of the International Bureau.

(4) Any country of the Union may, within a period of twelve months from the receipt of the notification, transmit its objections, if any, through the intermediary of the International Bureau, to the country or international intergovernmental organization concerned.

(5) In the case of State flags, the measures prescribed by [paragraph \(1\)](#), above, shall apply solely to marks registered after November 6, 1925.

(6) In the case of State emblems other than flags, and of official signs and hallmarks of the countries of the Union, and in the case of armorial bearings, flags, other emblems, abbreviations, and names, of international intergovernmental organizations, these provisions shall apply only to marks registered more than two months after receipt of the communication provided for in [paragraph \(3\)](#), above.

```
<?xml version="1.0" encoding="UTF-8" standalone="no" ?>
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            <content eid="art_1_para_1_content">
              <p>The countries to which this Convention applies constitute a union for the protection of industrial property.</p>
            </content>
          </paragraph>
          <paragraph eid="art_1_para_2">
            <num(2)></num>
            <content eid="art_1_para_2_content">
              <p>The protection of industrial property has as its object patents, utility models, industrial designs, trademarks, service marks, trade names, indications of source or appellations of origin, and the repression of unfair competition.</p>
            </content>
          </paragraph>
        </article>
      </body>
    </akoma>
```

- Multiple languages
- Timestamps
- Revision history
- Internal References
- External References
- Definitions

Example - Dates:

Was Country XY compliant with Treaty Z at DD-MM-YYYY in time?

Treaty



- Adoption date
- Entry Into Force
- Implementation Deadline
- Transition/Grace Period End

National Law



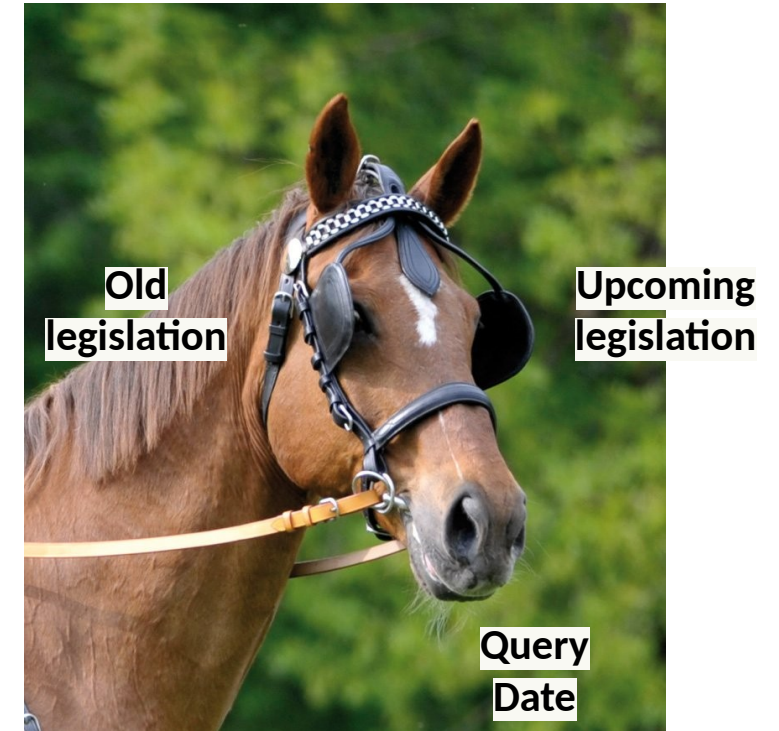
- Ratification date
- Entry Into Force
- Amendment Dates

Example - Dates:

Q: Was Austria compliant with Marrakesh on **February 20, 2019**?

Timeline:

— June 27, 2013	Treaty adopted
— Sept 30, 2016	Treaty enters into force (global)
— Oct 1, 2018	Austria ratifies
— Oct 11, 2018	EU implementation deadline
— Jan 1, 2019	Treaty enters into force for Austria
— Feb 20, 2019	Query date
— March 1, 2019	Austria amends Copyright Act



Example – Semantics

Copyright Length

Berne Convention (EN): "life of the author + 50 years"

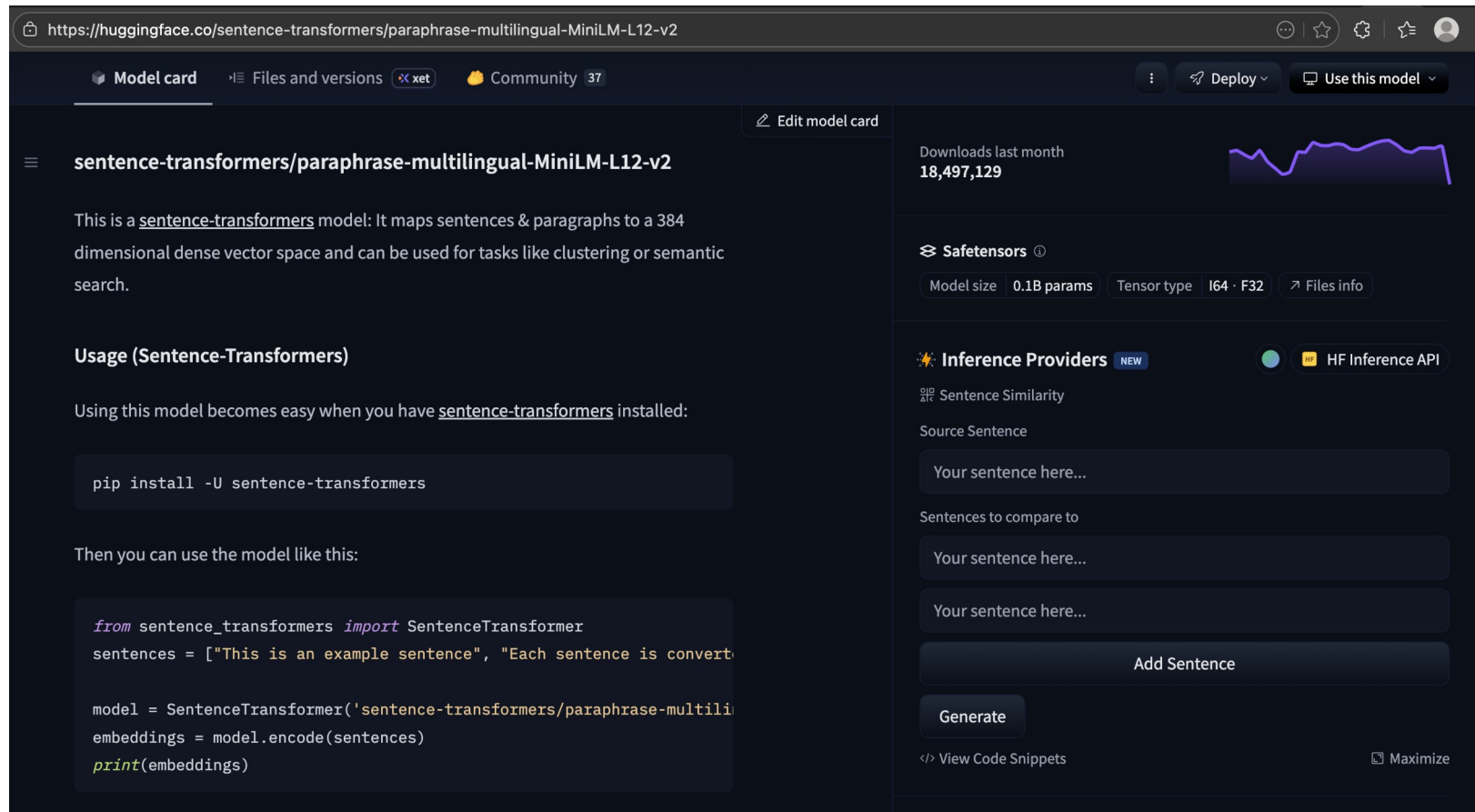
Swiss Law (DE): "70 Jahre nach dem Tod des Urhebers"

Italian Law (IT): "settanta anni dopo la morte dell'autore"

EU Directive (EN): "70 years after the author's death"

Is 'life + 70' compliant with a treaty requiring 'life + 50'?

Example – Semantics



The screenshot shows the Hugging Face model card for `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`. The page is in dark mode and includes a navigation bar with tabs for 'Model card', 'Files and versions', 'Community', 'Deploy', and 'Use this model'. The main content area on the left describes the model as a `sentence-transformers` model that maps sentences and paragraphs to a 384-dimensional dense vector space. It provides a usage example with Python code for installing the library and encoding sentences. The right sidebar shows the model's download statistics (18,497,129 downloads last month), safetensors information (0.1B params, I64 · F32), and inference providers (HF Inference API). It also features a 'Sentence Similarity' interface with input fields for source and comparison sentences, an 'Add Sentence' button, and a 'Generate' button.

<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

Model card | Files and versions | Community 37 | Deploy | Use this model

sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2

This is a `sentence-transformers` model: It maps sentences & paragraphs to a 384 dimensional dense vector space and can be used for tasks like clustering or semantic search.

Usage (Sentence-Transformers)

Using this model becomes easy when you have `sentence-transformers` installed:

```
pip install -U sentence-transformers
```

Then you can use the model like this:

```
from sentence_transformers import SentenceTransformer
sentences = ["This is an example sentence", "Each sentence is converted to an embedding"]

model = SentenceTransformer('sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2')
embeddings = model.encode(sentences)
print(embeddings)
```

Downloads last month: 18,497,129

Safetensors

Model size: 0.1B params | Tensor type: I64 · F32 | Files info

Inference Providers | NEW | HF Inference API

Sentence Similarity

Source Sentence: Your sentence here...

Sentences to compare to: Your sentence here... | Your sentence here...

Add Sentence

Generate

View Code Snippets | Maximize

Example – Semantics

 **Inference Providers** NEW

HF Inference API

 Sentence Similarity

Source Sentence

Your sentence here...

Sentences to compare to

Your sentence here...

Your sentence here...

Add Sentence

Generate

</> View Code Snippets

 Maximize

 **Inference Providers** NEW

HF Inference API

 Sentence Similarity Reset

Source Sentence

70 Jahre nach dem Tod des Urhebers

Sentences to compare to

settanta anni dopo la morte dell'autore

70 years after the author's death

70 Jahre nach dem Tod des Urhebers

Add Sentence

Generate

settanta anni dopo la morte dell'autore

0.816

70 years after the author's death

0.838

70 Jahre nach dem Tod des Urhebers

1.000

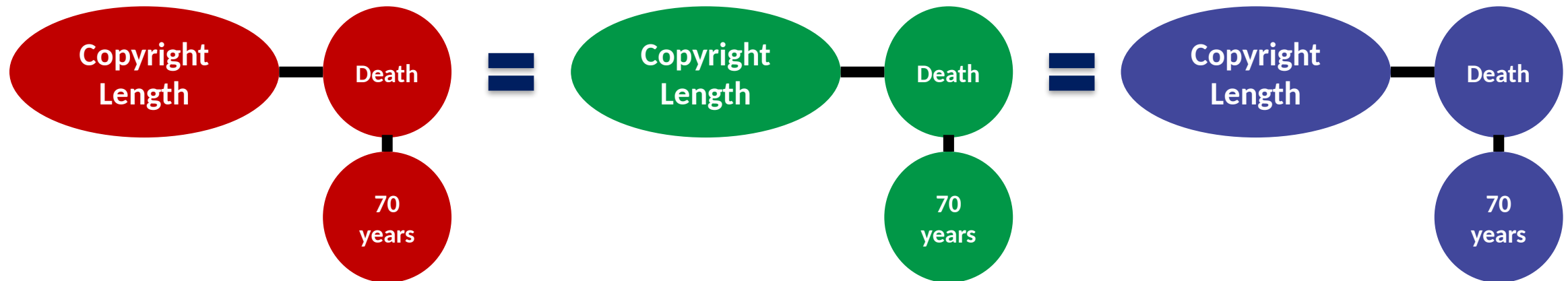
</> View Code Snippets

0.3s

 Maximize

Example – Semantics

Solution: Country-specific Knowledge Graphs



Example – Cross References: